MTH 9897 SYSTEMATIC TRADING | PROJECT 8

The Conservative Formula

A replication using CRSP data, 1929-2024

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Reference: David Blitz and Pim van Vliet (2018), The Conservative Formula: Quantitative Investing Made Easy, The Journal of Portfolio Management 44(7); working paper SSRN 3145152.

Abstract

We implement the Conservative Formula, a quarterly portfolio of 100 stocks selected using volatility, momentum and net payout. Over 1929-2016, our portfolio earns 14.80% compounded annually with 17.06% volatility, versus the paper's 15.10% and 16.50%. We then extend the data through 2024, account for costs, and test whether the return can be explained by familiar equity factors.

In 1990-2024, net of 30 bps per dollar traded, annual Carhart alpha is 0.20%, with a 95% interval [-1.98%, 2.39%]. The interval includes zero. We obtain a close historical replication, but recent factor-adjusted performance is less convincing.

The assignment highlights two possible implementation mistakes: using dividends alone for net payout, and overlooking sector positions. We compare dividends with net payout, measure industry contributions, and test a sector-by-sector selection rule. We also examine long/short trading, linear and quadratic optimization, and a rank buffer intended to reduce turnover.

Scope	Evidence
Paper replication	1929-2016; 1,056 monthly observations
Extended research	1929-2024; 1,152 monthly observations
Implementation	Monthly CRSP inputs; quarterly baseline; separate signal and trading experiments

1. What we test

The appeal of the paper is that its rule is easy to describe and can be implemented with monthly prices, returns and shares. The Conservative portfolio combines low historical return volatility with strong price momentum and high net payout. Its Speculative mirror uses the high-volatility half and weak combined scores.

Question	Test and interpretation
Can the paper be reproduced?	Compare gross return, risk, turnover, control portfolios and regression alphas on 1929-2016 under the paper's definitions.
Is return explained by known factors?	Carhart four-factor alpha is primary. Fama-French five factors plus momentum is a robustness check from July 1963.
Has performance weakened?	Fixed subperiods and rolling 120-month estimates. The recent comparison is 1990-2024; 2017-2024 follows the paper's sample.
What survives costs?	Returns at 10 and 30 bps per dollar traded and break-even costs for market excess return and Carhart alpha.

Comparisons and uncertainty

We use 1929-1962, 1963-1989, 1990-2016 and 2017-2024, plus the combined recent period 1990-2024. The final period contains only 96 months. Newey-West intervals account for serial dependence; the portfolio regressions use 12 lags. Some performance intervals use moving-block bootstrap resampling. Overlapping diagnostic horizons use longer lag choices recorded in their result tables.

Annual compound return is (product of 1 + monthly return) raised to 12/T, minus 1. Annual arithmetic return is 12 times the monthly mean; volatility is the monthly sample standard deviation times the square root of 12. These measures are kept distinct. Information ratios divide annualized active return by tracking error; the long/short exercise instead reports excess over collateral risk-free return.

We fixed the primary comparison periods before computing post-2016 portfolio returns and fixed the rank-buffer rule before testing that variant. The additional payout/sector comparisons, FF3 stock-beta controls and quadratic model came after the baseline analysis. We therefore treat them as exploratory; the already-observed 2017-2024 period is not a holdout for them.

This report covers the US assignment. It does not reproduce the paper's international FactSet samples, and CRSP alone does not supply the book-equity inputs needed for a separate Value control.

2. Data and cleaning

We use full CRSP legacy web exports from WRDS, with monthly history from December 1925 through December 2024. The normalized panel contains 3,954,591 security-months and 26,746 securities. There are 3,747,988 eligible observations. Source receipts, query metadata, input counts and SHA-256 hashes identify the extract; a CRSP release identifier was not provided.

Input	Use
Monthly stock file	Total/price returns, price, shares, monthly volume and split factors
Stock names/history	Time-varying share codes, exchanges, company identifiers and SIC industries

Input	Use
Delisting file	Terminal event dates, codes and delisting returns
Distribution file	Independent checks of implied cash dividends and corporate actions
Monthly index file	CRSP value-weighted benchmark and total market size for capacity normalization
FF3, momentum and FF5	Risk-free rate and factor-attribution regressions

Cleaning choices

We keep common-share codes 10/11 and exchange codes 1/2/3. Each month we retain the largest common share class per company to avoid double counting. Historical names are joined at their effective dates. Terminal events and delisted securities remain in the holding-period returns; later rows following the terminal event are removed. There are no duplicate security-months or duplicate eligible company-months.

Data convention	Count or rule
Terminal events	22,869: 20,022 observed returns; 2,530 imputed; 317 set to zero
Performance delisting imputation	-30% for NYSE/AMEX and -55% for Nasdaq where the documented rule applies
Missing held returns	Stale-price carry or an in-gap price ratio; 447 stale holding months in the audited universe
Historical universe coverage	393 names at December 1928; first full 1,000-name universe at 1964-06-30

Missing returns remain missing in historical signal windows. The held-return convention is separate: it keeps a position rather than silently dropping a missing holding observation. The dividend and delisting audits document the limits of these conventions, especially in early data.

3. Signals and portfolio rules

Point-in-time signal definitions

```
Adjusted price P[t] = abs(PRC[t]) / CFACPR[t]
Adjusted shares S[t] = SHROUT[t] * CFACSHR[t]
Volatility = sd(RET_total[t-35], ..., RET_total[t]) * sqrt(12)
Momentum = product(1 + RET_price[t-11], ..., RET_price[t-1]) - 1
Dividend yield = sum(D_adj[t-11], ..., D_adj[t]) / P[t]
Net payout = dividend yield + 1 - S[t] / mean(S[t-23], ..., S[t])
```

Price momentum contains 11 monthly returns and excludes the latest month. Volatility uses 36 consecutive monthly total returns. Implied adjusted cash dividends use total return minus price return times the preceding adjusted price. The 12-month dividend window and 24-month share average including the formation month are disclosed implementation conventions.

Order	Quarter-end action
1	Apply eligibility, positive capitalization, non-terminal status and 36-month history.
2	Choose the largest 1,000 eligible stocks by security capitalization, or all if fewer exist.

Order	Quarter-end action
3	Retain the lower-volatility half; then require complete momentum and net payout signals.
4	Rank momentum and net payout favorably within the pool; average the two ranks.
5	Select the best 100, resolve ties deterministically, and assign 1% weights.
6	Hold the next three months; weights drift, with terminal proceeds held as cash earning RF.

We apply the history requirement before the size cut, following the paper. Selection uses information available at the quarter-end before the first holding month. Signals do not change when later input values are changed in the tests. The paper compares ranks across industries, so we keep that rule in the original portfolio and examine industry constraints separately.

Trading and costs

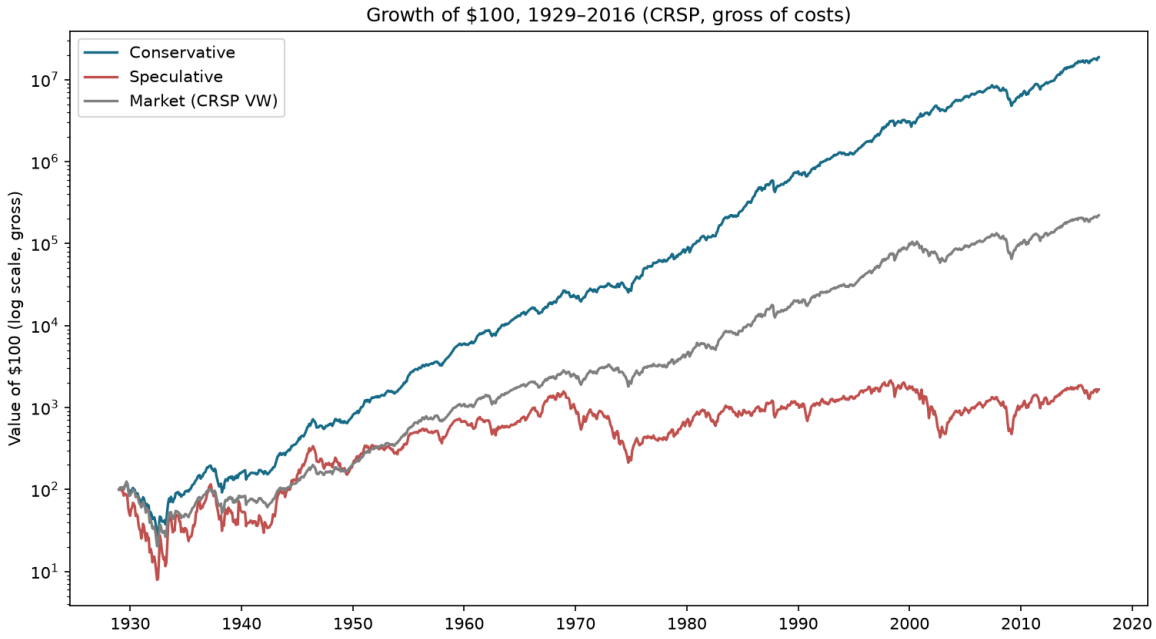
Trading cost is charged on total absolute changes in stock weights, with initial investment included in NAV. Published quarterly turnover is single-counted, one half of this traded notional, excluding the initial allocation. At a flat cost c, net monthly return is (1 - c times traded notional) times (1 + gross return), minus 1. This timing differs slightly from subtracting an annual cost estimate from CAGR.

4. How close are we to the paper?

The Conservative baseline earns 14.80% gross annually with 17.06% volatility over 1929-2016. The market earns 9.15%; the Speculative control earns 3.24% with 35.87% volatility. The Conservative/Speculative comparison has the same broad pattern as the paper, although several numbers differ.

Portfolio	Paper CAGR	CRSP CAGR	Paper vol.	CRSP vol.
Formula	15.10%	14.80%	16.50%	17.06%
Market	9.30%	9.15%	18.70%	18.70%
Small	12.00%	12.95%	31.40%	33.47%
Momentum	15.90%	15.49%	25.70%	26.10%
Lowvol	10.20%	10.41%	13.30%	14.42%
NPY	14.50%	14.18%	20.10%	21.15%

Gross returns; 1,056 months. Value is excluded because book equity is unavailable in the CRSP-only inputs.



Growth of $100 using monthly gross returns. A logarithmic scale makes long-sample compound growth comparable.

Turnover is the largest unresolved implementation difference. Quarterly single-counted turnover is 38.9%, versus 32.0% in the paper. We cannot tell how much of the gap comes from ranking inputs and how much comes from the authors' turnover convention. We leave it visible in the comparison.

5. What the share-count term adds

Dividend yield misses the indirect payout from share repurchases, which the assignment specifically asks us to account for. A reduction in adjusted shares raises the payout signal; net issuance lowers it. A split must not alter the adjusted share count. With dividend yield of 3%, 90 current adjusted shares and a 100-share average, net payout is 13%.

Strategy	Gross CAGR	Net 30 CAGR	Quarterly turnover
Baseline NPY	14.80%	13.73%	38.91%
Dividends-only counterfactual	13.44%	12.39%	38.62%
Sector quota variant	14.51%	13.37%	41.75%

The paired net-return difference between the original NPY rule and dividends alone is 1.22% per year in arithmetic terms, with a Newey-West 95% interval [0.56%, 1.87%]. Universe, low-volatility pool, complete-case membership, momentum, portfolio size, weights, dates and costs are matched; only the payout ranking is replaced.

Period	Dividend yield	Net share proxy	NPY	Overlap with dividends-only
1929-2016	5.07%	1.53%	6.60%	65.9%
2017-2024	2.00%	3.53%	5.53%	52.0%
1990-2024	2.42%	3.32%	5.74%	49.9%

Formation averages among baseline holdings; these are signal components, not realized cash payout returns.

Checking the payout calculation

Synthetic cases check buybacks, issuance and split neutrality. Direct recalculation on 1,200 actual observations verifies the 36-, 12- and 24-month windows, with maximum differences below 8.0e-15. Implied dividends are separately compared with CRSP distributions; approximately 96.8%-99.85% of checkable ordinary-cash cases reconcile by decade.

Adjusted share changes capture net repurchases after issuance; they are not observed buyback cash expenditures. Other corporate actions can also change shares. Dividend timing, special distributions and incomplete observations remain sources of uncertainty. The matched comparison shows a material signal difference, not a causal estimate of buybacks.

6. Industry weights and stock selection

We compare the original portfolio with a capitalization-weighted reference drawn from the same formation universe. Constituents and industry labels are fixed for each quarter; both portfolios drift with returns and move terminal proceeds to cash. This avoids confusing a broad-market universe difference with industry allocation.

```
Baseline - CRSP market
  = industry allocation + stock selection
    + reference universe - CRSP market
Allocation = sum[(w_portfolio - w_reference) * r_industry]
Selection = sum[w_portfolio * (r_portfolio_ind - r_industry)]
```

The stock-selection term includes the allocation/selection interaction. The identity is reconciled monthly, including cash. Annual contributions are 12 times the monthly mean, rather than contributions to compound return.

Period	Allocation	Selection	Universe bridge	Total excess
1929-2016	0.82%	3.80%	0.18%	4.80%
2017-2024	-1.54%	-2.06%	1.86%	-1.74%
1990-2024	-0.04%	1.08%	0.66%	1.69%

Annual arithmetic contributions. They describe portfolio positioning and do not identify causal industry effects.

Selecting stocks within each industry

Following the professor's Enhanced Momentum example, the variant allocates 100 slots in proportion to the universe's FF12 industry stock counts. Largest-remainder rounding sets integer quotas. Within each industry it retains the low-volatility half, then ranks momentum and net payout and fills its quota. Stocks receive equal weights.

All 384 formations hold 100 names. The largest deviation from an industry's count-proportional target is 0.70 percentage points. The variant earns 14.51% gross annually over 1929-2016 versus 14.80% for the original baseline, with more turnover. These values are also shown in the payout comparison in Section 5.

Industry neutrality is not silently imposed on the paper baseline. The quota variant changes the global volatility screen and is exploratory. FF12 industries are the available US grouping used for both the diagnostics and these sector-positioning checks.

7. How much return is explained by factors?

Monthly portfolio excess returns are regressed on the market factor, then size/value, momentum and the FF5 profitability/investment factors. Carhart is the primary attribution model. FF5 plus momentum uses July 1963 onward, so its result is not a full 1929-2016 estimate.

Portfolio	Model	Annual alpha	NW t	Months
Conservative	CAPM	5.80%	6.41	1056
Conservative	Carhart	3.23%	4.63	1056
Conservative	FF5+UMD	-0.11%	-0.14	642
CMS	CAPM	12.33%	6.13	1056
CMS	Carhart	7.74%	5.34	1056
CMS	FF5+UMD	3.12%	2.43	642

CMS is the Conservative minus Speculative contrast. Gross rows compare with the paper; confidence inference uses Newey-West errors.

The Conservative portfolio's FF5-plus-momentum alpha is -0.11% (NW t -0.14). The reduction in alpha when profitability and investment are included is consistent with the formula capturing known factor exposures, rather than creating an independent source of return.

Replicating the paper's regression tables

CMS comparison	Sample	Paper alpha	CRSP alpha
A: CAPM	1929-01..2016-12	13.90%	12.33%
A: Carhart	1929-01..2016-12	8.80%	7.74%
B: CAPM	1963-07..2016-12	14.00%	14.20%
B: FF3	1963-07..2016-12	13.00%	12.99%
B: FF5	1963-07..2016-12	8.80%	8.69%
B: FF5 + MOM	1963-07..2016-12	3.30%	3.12%

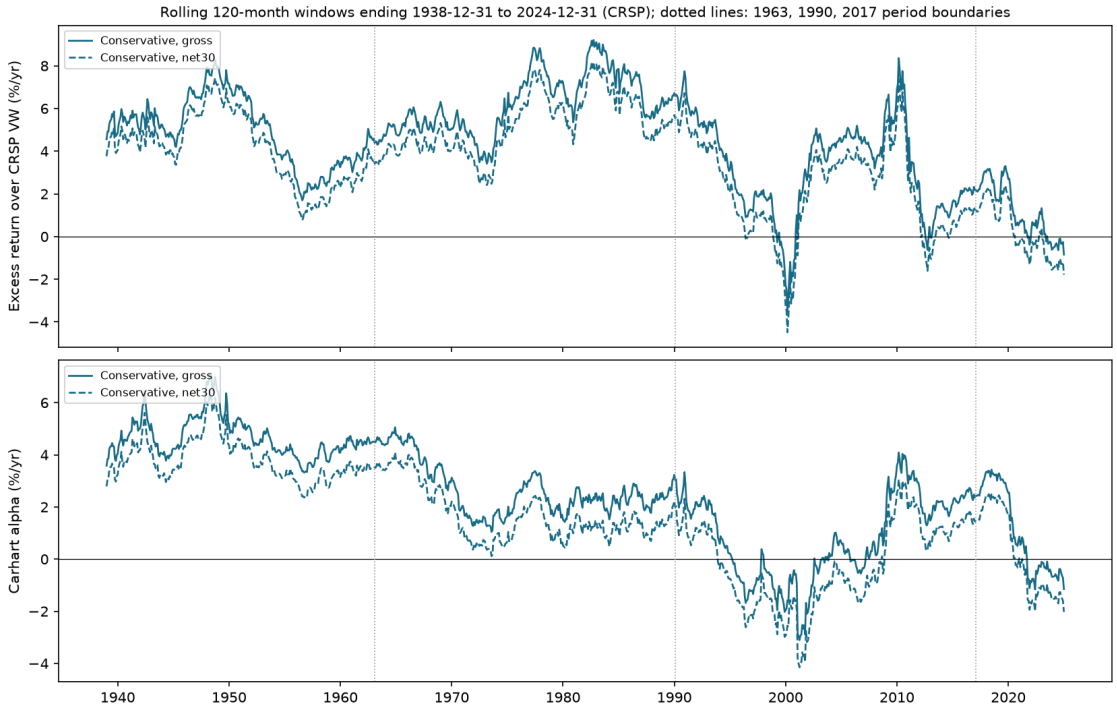
Full-sample Panel A alphas differ materially; the post-1963 Panel B comparisons are closer. A gross long/short difference is not the same as an investable cost-inclusive spread. The result tables distinguish cost-inclusive CMS from the difference of two net long-only return series.

8. Costs and changes over time

Period	Net CAGR	Excess vs market	Carhart alpha	95% alpha interval
1929-1962	12.96%	4.33%	3.80%	[2.00%, 5.60%]
1963-1989	16.96%	5.36%	1.16%	[-0.29%, 2.62%]
1990-2016	11.53%	1.74%	0.73%	[-1.80%, 3.25%]
2017-2024	9.97%	-2.70%	-2.10%	[-5.99%, 1.80%]
1990-2024	11.17%	0.73%	0.20%	[-1.98%, 2.39%]

Flat 30 bps per dollar traded. Excess and alpha are annual arithmetic measures, distinct from compound return.

Net 30-bps CAGR is 13.73% over the paper sample. The recent alpha break-even cost is 36 bps per dollar traded. For 2017-2024, gross excess return and alpha are already nonpositive, so no positive trading-cost threshold restores them. High early-period break-even estimates do not establish that historical trading was cheap.



Rolling 120-month excess return and Carhart alpha; gross and net series use the same holdings. Window estimates are descriptive and strongly overlapping.

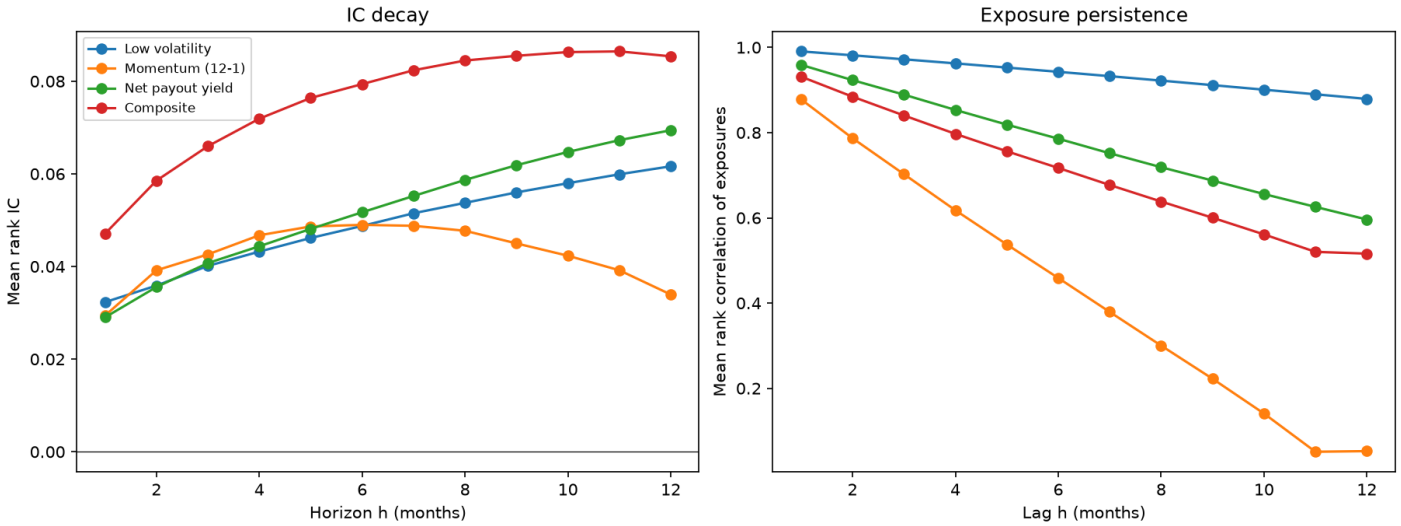
The recent net alpha interval includes zero. Positive and negative expected alpha both remain plausible, particularly in the short sample after 2016.

9. Examining each signal

Diagnostics use the monthly history-first universe. Low volatility is oriented favorably as negative volatility. Each signal is demeaned within industry, divided by the cross-sectional standard deviation of demeaned values, and clipped to [-3, 3]. Groups with fewer than five members join Other. The diagnostic composite is the mean of the three complete neutralized exposures; it differs from the paper's rank-and-screen rule.

Forward total returns retain the formation cohort and terminal events. Industry adjustment subtracts the equal-weight return of the same formation industry cohort. Diagnostic cash after an exit earns zero, a disclosed convention distinct from RF cash in the traded baseline. IC is the cross-sectional Spearman correlation with forward industry-adjusted returns.

One-month exposure	Mean IC	IC variance	Mean/SD	NW t
Low volatility	0.0323	0.0335	0.18	5.42
Momentum	0.0294	0.0226	0.20	6.75
Net payout	0.0290	0.0123	0.26	8.01
Composite	0.0471	0.0270	0.29	8.76



IC decay with holding horizon and persistence of the exposures. Overlapping horizons use lag-aware inference.

Composite D10 minus D1, one month	Mean adjusted spread	NW t
1929-1962	-0.03%	-0.06
1963-1989	1.02%	4.31
1990-2016	0.68%	1.98
2017-2024	0.40%	0.66

The notebook additionally shows moving IC, decile stock hit rates, all horizons and literal mean/standard-deviation ratios.

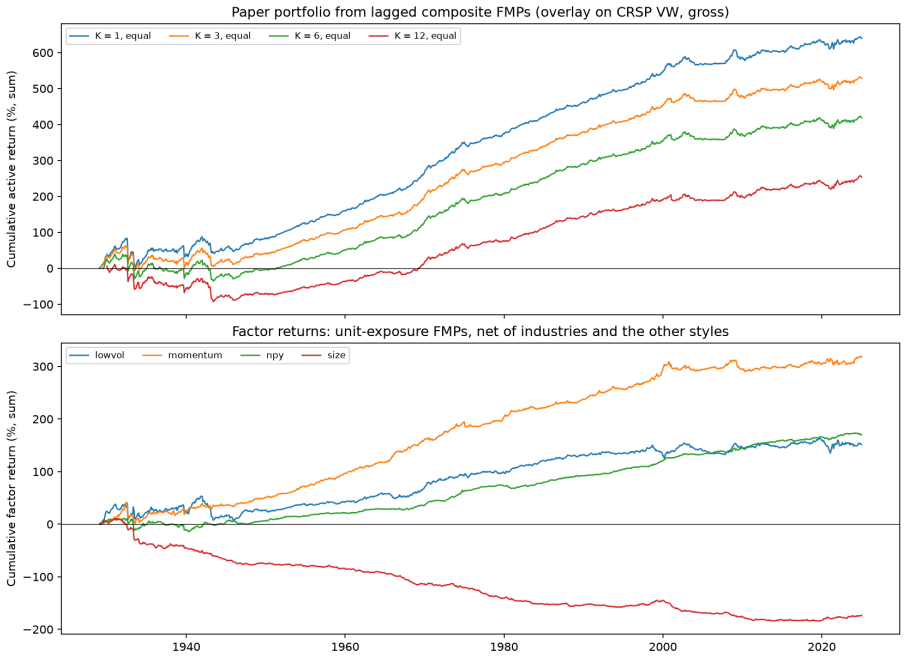
10. Factor-mimicking portfolios

Weighted least squares includes the three diagnostic signals, log-size exposure, industry dummies and historical FF3 stock loadings. The cross-sectional WLS observation weight is the square root of market capitalization. The regression operator maps security returns to unit-exposure factor portfolios. The composite targets unit exposure to each of the three signal factors and zero exposure to the risk controls.

Stock FF3 betas use up to 36 monthly excess returns ending at formation, with at least 24 factor-covered observations and an intercept. All 1,152 formation checks use historical returns; the minimum factor coverage is 30

months. We added these controls after the baseline analysis; only historical stock returns are used to estimate them.

Lagged portfolios K	Annual active mean	IR	Monthly turnover	Gross leverage
1	6.66%	0.59	46.74%	2.52
3	5.51%	0.53	24.96%	2.45
6	4.38%	0.44	17.09%	2.38
12	2.67%	0.29	11.41%	2.26



Lagged composite paper portfolios. Equal and exponentially decaying weights test the persistence/turnover tradeoff.

These are gross diagnostic factor portfolios with signed weights and potentially substantial leverage. Cumulative active performance is not a funded long-only NAV. Costs and borrow are explicitly modeled in the separate trading exercise; the WLS result is not an implementable net strategy claim.

11. Long/short trading

The simple trading exercise buys the top and sells the bottom composite decile. Cohorts are added monthly and expire after n months, for n = 1, 3, 6 or 12. Costs of 0, 5, 10, 20 and 30 bps per dollar traded and size-dependent monthly borrow costs are compared. Trades, positions, NAV accounting and long/short contributions are recorded.

Holding n	Annual excess over RF	IR	Max drawdown	Market corr.
1	-0.98%	-0.04	-99.95%	-0.61
3	0.64%	0.03	-99.97%	-0.60
6	-0.02%	-0.00	-99.99%	-0.60
12	-1.86%	-0.08	-99.99%	-0.59

Monthly-reweighted alternative, 1929-2024. Values rounded to -100% indicate extremely deep drawdown, not necessarily an exact zero NAV.

The original buy-and-hold cohorts with n at least 3 are ruined in the early sample, including August 1932 and May 1933. They are retained and labeled as failed rather than continued with fictitious positive wealth. Monthly reweighting was introduced after that outcome and is explicitly a subsequent convention; its severe drawdowns also limit the economic attractiveness of the extension.

n = 3 contribution	Annual arithmetic amount
Long positions	11.82%
Short positions	-11.17%
Borrow cost	2.16%
Collateral RF	3.23%

How large could the portfolio be?

The n = 3 monthly-reweighted capacity estimate for the 2020s is $7.93 billion at the 10% volume assumption. Holdings and turnover are related to reported monthly dollar volume. The estimates use a high participation quantile and normalize historical AUM to December 2024 market size, not CPI-adjusted dollars. They omit market impact, daily trade timing and actual borrow availability; they are not executable AUM guarantees.

12. Linear optimization

The long-only linear optimizer maximizes a signal-weighted portfolio subject to full investment, nonnegative weights, a 2% stock cap, industry bands around universe capitalization weights, and a two-way turnover budget. The specified objective uses the industry-neutral composite z score. Transaction costs scale upward for smaller or more volatile stocks, with base levels of 10 or 30 bps.

```
Maximize: sum(alpha[i] * w[i]) - sum(c[i] * abs(w[i]-w_prev[i]))
Subject to: sum(w)=1; 0 <= w[i] <= 0.02
            industry lower <= industry weight <= upper
            sum(abs(w-w_prev)) <= turnover budget
```

Industry band	Two-way budget	Net IR	Carhart alpha	95% interval
2%	20%	0.42	1.64%	[0.27%, 3.01%]
2%	40%	0.44	1.78%	[-0.13%, 3.69%]
5%	20%	0.38	1.31%	[-0.05%, 2.67%]
5%	40%	0.48	1.95%	[0.15%, 3.75%]

For these four recent-period examples, two of the four 95% intervals exclude zero. They are nominal intervals selected from a broader parameter grid and are not adjusted for multiple comparisons. The 2017-2024 grid produces no alpha interval excluding zero among the 12 specified 30-bps examples. Positive selected-grid estimates are not independent validation.

Expected-return sensitivity and units

A separate exploratory objective converts a signal to expected three-month return using only completed trailing IC estimates: alpha = IC times annual volatility times sqrt(3/12) times exposure. The sqrt(3/12) conversion reflects the annual volatility units; sqrt(3) would incorrectly treat that input as monthly. The original z-score objective is unchanged.

When a stock cap makes an industry's lower band infeasible, the relaxation is recorded. Minimum feasible turnover is determined separately; any needed budget relaxation is reported rather than hidden. Capacity, active skew/kurtosis, market correlation, hit rates and drawdowns are included in the grid result files.

13. Adding a covariance penalty

We also add a portfolio covariance penalty to expected return and trading cost. The risk model uses the last 36 monthly total returns ending at formation. It combines 50% sample covariance with 50% diagonal covariance and scales monthly covariance to three months, assuming no serial correlation.

```
Maximize: alpha' w - gamma/2 * w' Sigma_3m w
          - sum(c[i] * abs(w[i]-w_prev[i]))
Sigma_3m = 3 * (0.5 * sample_cov + 0.5 * diagonal_cov)
alpha = completed trailing IC * annual vol * sqrt(3/12) * z
```

Parameters are fixed heuristics: risk aversion gamma = 5 or 20; 2% stock caps; industry bands of +/-5 percentage points; 40% two-way quarterly turnover after initial allocation; and scaled costs starting at 30 bps. Until enough completed IC history exists, expected alpha is zero. The same formation and holding conventions apply as in the baseline.

Period / gamma	Net CAGR	Volatility	IR	Carhart alpha [95% CI]
2017-2024 / 5	11.26%	14.13%	-0.27	-1.28% [-5.42%, 2.87%]
1990-2024 / 5	11.59%	12.51%	0.14	1.38% [-0.32%, 3.08%]
2017-2024 / 20	9.93%	13.20%	-0.45	-1.58% [-4.90%, 1.74%]
1990-2024 / 20	10.39%	11.51%	-0.03	1.09% [-0.65%, 2.83%]

Solver verification. Clarabel returns Solved for all 768 formations across the two variants. The maximum independently measured constraint violation is 4.91e-09. There are no risk-window lookahead violations and no turnover-budget relaxations. Five infeasible industry lower bounds per variant are explicitly relaxed and recorded.

An independent dense constrained-optimization test verifies the quadratic objective. Market correlations remain high, about 0.87-0.89 in the recent sample. The model is long-only and industry-constrained; lower volatility does not imply market neutrality. The extension was introduced after seeing baseline results and does not establish an optimal parameter choice.

14. Reducing turnover with a rank buffer

The rank-buffer variant retains an existing holding while it remains among the best 150 ranks in the eligible low-volatility pool, then fills vacancies with the best available names. It targets 100 equally weighted holdings quarterly. The documented success rule asks for at least 25% less turnover and a positive net-return difference, rather than a higher gross backtest selected after the result.

Buffer minus baseline	Annual paired difference	95% interval	NW t
gross	0.06%	[-0.37%, 0.50%]	0.29
net10	0.16%	[-0.28%, 0.59%]	0.70
net30	0.34%	[-0.10%, 0.78%]	1.52

Quarterly turnover falls from 39.8% to 28.4%. The observed net difference can reflect both a changed gross path and lower trading cost. A small observed gross difference does not prove that the signal's economic quality is unchanged. The paired intervals remain important even when the mechanical success criterion passes.

Checking the calculations

Pass	Evidence
Paper and assignment	Construction order, momentum timing, payout definition, sector distinction and all required diagnostics/construction exercises.
Data and accounting	18 raw/normalized/audit input hashes; duplicates; splits, dividends, delistings, missing returns and company classes.
Independent calculations	38,400 selected names; 1,152 gross/net monthly returns from a separate dollar-holdings recursion; direct signals, Carhart/Newey-West and sector reconciliation.

All 13 independent accounting checks pass. The complete scientific test suite passes 139 tests. The executed notebook runs 16 code cells without errors in empirical and synthetic modes; a fresh checkout also runs the empirical report without licensed data. Cache hashes and producing data/source/package identities are checked before reuse. This validates the tested implementation; it does not remove the data and modeling assumptions.

15. Running the analysis

Main calculation entry point: `research/run_all.py`, run as a module from Final. It runs all 12 studies and saves their results and file hashes. The Jupyter notebook is the primary assignment deliverable.

File or module	Role
research/run_all.py	Runs the full research pipeline or selected studies.
crsp_pipeline	Imports or extracts raw data, normalizes the panel, writes audits.
research/engine.py	Baseline selection and portfolio accounting.
build_notebook.py / execute_notebook.py	Generate and execute the Jupyter report.
build_report_pdf.py	Generate this written PDF from verified aggregate results.

File or module	Role
presentation/build_pdf.py	Generate the separate 18-slide PDF and notes.
package_submission.py	Create the private runnable package with code and data.

Complete rebuild with the private inputs

```
cd Final
python -m pip install -r requirements.txt
python -m crsp_pipeline build
python -m research.run_all
python build_notebook.py
python execute_notebook.py
python build_report_pdf.py
python presentation/build_pdf.py
python -m pytest tests -q
python package_submission.py
```

The full reviewed research rebuild took approximately 16 minutes on the tested Windows environment; machine and library versions affect runtime. Cached notebook, written PDF and slides generate in seconds from the supplied aggregate results. Dependencies are recorded in requirements.txt and the tested-version lock file.

Submission: include the executed notebook, code and data required by the assignment. This report and the slide PDF are companions. The private ZIP includes the seven raw parquet inputs, normalized inputs and worked examples. Licensed CRSP rows stay in the authorized course package; GitHub contains source and aggregate outputs.

16. Conclusions

Our implementation reproduces the main contrast between Conservative and Speculative stocks: higher compound return with lower realized volatility. The net-payout share-count term materially changes selections and returns relative to dividend yield alone. Industry attribution shows that both positioning and stock selection contribute, and the professor's sector-quota construction is tested explicitly.

We are less confident about recent outperformance. After factor adjustment and costs, the original portfolio's recent Carhart alpha is not significantly different from zero. The optimizers and buffer change risk and turnover, but their results do not yet give us evidence of a better strategy on an unseen sample. Lower volatility is the more consistent finding.

Limitation	Implication
Turnover 38.9% versus paper 32%	An unresolved implementation/definition gap remains.
Early universe below 1,000	The 1929-1964 strategy uses all available history-eligible names.
Share-class and size conventions	Security capitalization and largest-class selection may differ from the authors' universe.
Dividend and terminal-return proxies	Corporate-action timing and imputation introduce model uncertainty.
Some regression/control mismatches	Close return/risk replication does not imply all paper tables are matched.

Limitation	Implication
Short post-2016 period and grid selection	Unadjusted intervals and exploratory choices limit inference.
Capacity and trading assumptions	Monthly volume omits impact, trade scheduling and borrow availability.

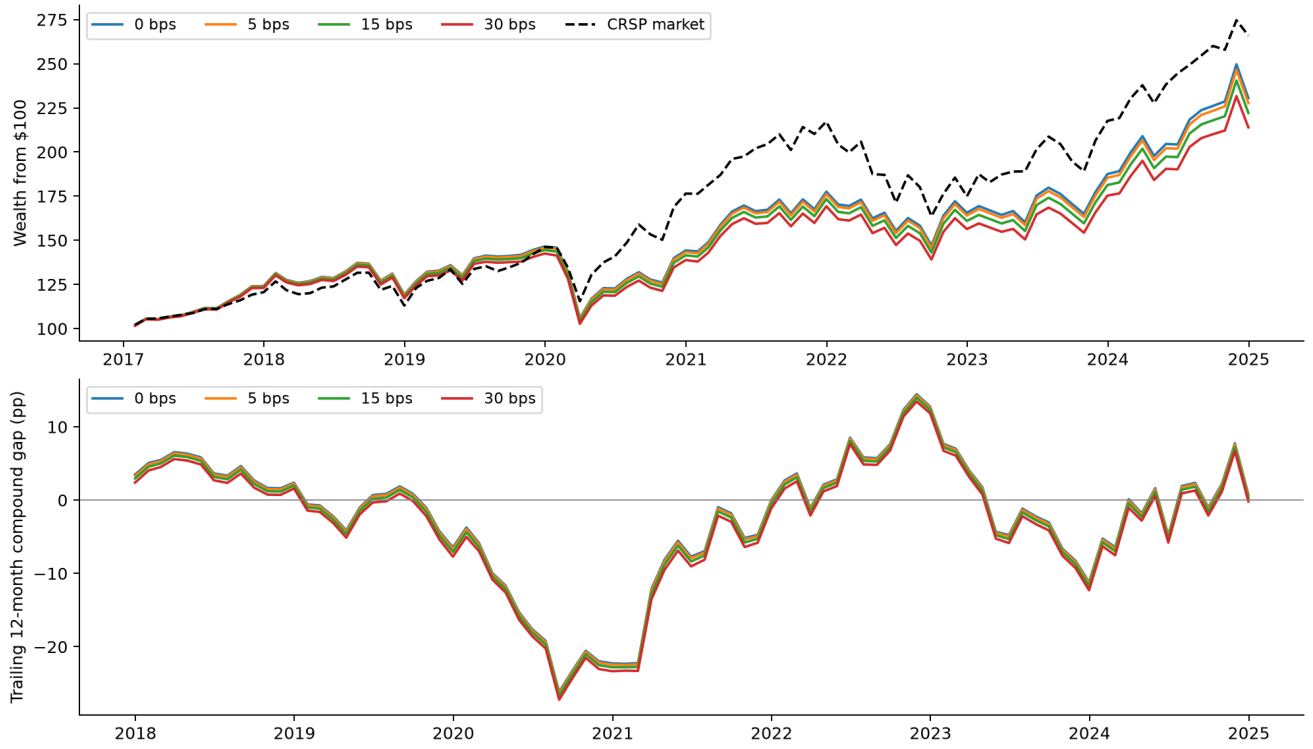
17. Follow-up to presentation feedback

The professor suggested a finer recent-period cost grid, costs that change with average spreads, and a UK comparison. We keep the original US stock-selection rule and report these follow-ups separately. The transcript does not specify a forecasting model. Our time-series analysis therefore examines annual returns, rolling return gaps and rolling factor alpha.

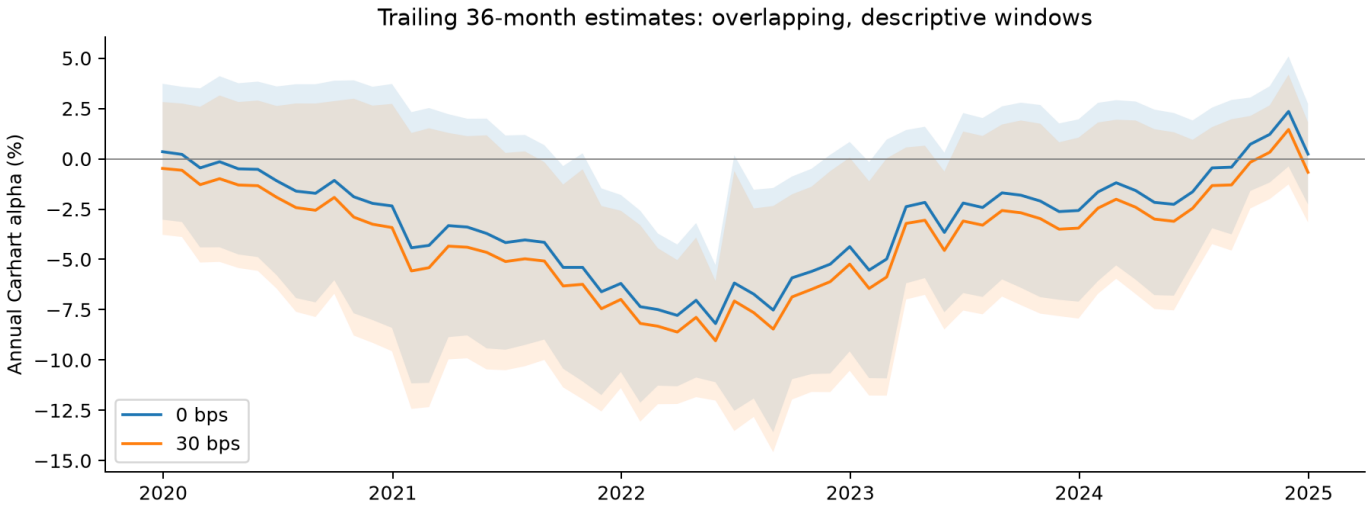
US costs, 2017-2024

Cost (bps)	Annual return	Above market / yr	Carhart alpha / yr	Final $100
0	11.01%	-1.74%	-1.21%	$230.55
5	10.83%	-1.90%	-1.36%	$227.69
10	10.66%	-2.06%	-1.51%	$224.86
15	10.49%	-2.22%	-1.66%	$222.08
20	10.32%	-2.38%	-1.80%	$219.32
25	10.14%	-2.54%	-1.95%	$216.60
30	9.97%	-2.70%	-2.10%	$213.91

The table charges every dollar purchased or sold, including the initial investment. Annual return is compounded; market excess is 12 times the average monthly strategy-minus-CRSP-market return. Carhart alpha is annualized from a monthly US four-factor regression with Newey-West 12-lag intervals. All seven recent alpha intervals contain zero. Positive absolute profit coexists with market underperformance even before costs.



Recent wealth paths and overlapping trailing 12-month return gaps.



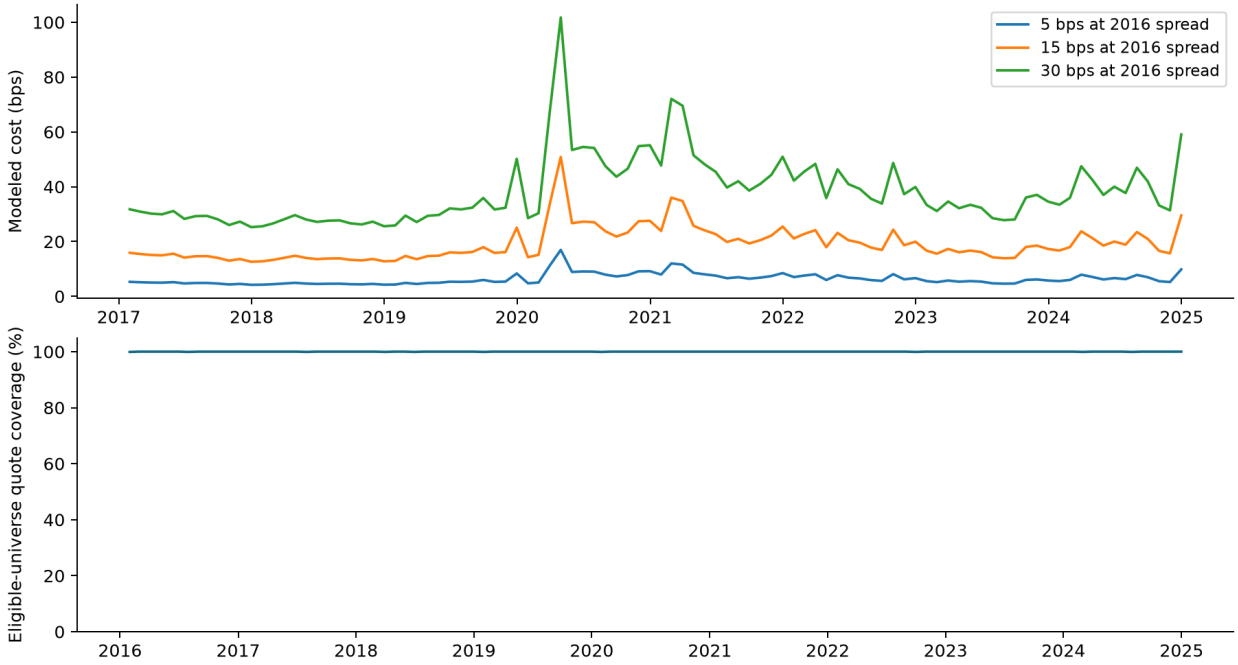
Trailing 36-month Carhart alpha and HAC 95% intervals. Overlapping windows are descriptive, with no formal break-date or forecasting claim.

Lagged spread scaling

We use actual CRSP closing BID and ASK fields, not intraday high and low. After checking positive and ordered quotes, we compute the quoted half spread as 5,000 x (ASK-BID) / midpoint in bps. Full relative spreads above 50% are excluded. Each month averages the largest 1,000 eligible history-qualified stocks. Coverage must exceed 80% in every month.

For each reference level c in 0, 5, ..., 30 bps, cost in month t is c times the prior calendar-month average half spread divided by the fixed 2016 average. This uses only information available before the return month. The reference level describes 2016 conditions rather than the realized 2017-2024 average. Costs affect rebalance trades only. Closing quotes measure relative liquidity, not actual execution costs.

2016 reference bps	Mean modeled bps	Annual return	Market excess / yr
0	0.00	11.01%	-1.74%
5	6.35	10.79%	-1.94%
10	12.71	10.57%	-2.15%
15	19.06	10.35%	-2.35%
20	25.41	10.14%	-2.55%
25	31.77	9.92%	-2.75%
30	38.12	9.70%	-2.95%



Lagged quote-scaled costs and monthly quote coverage.

UK sample and cleaning

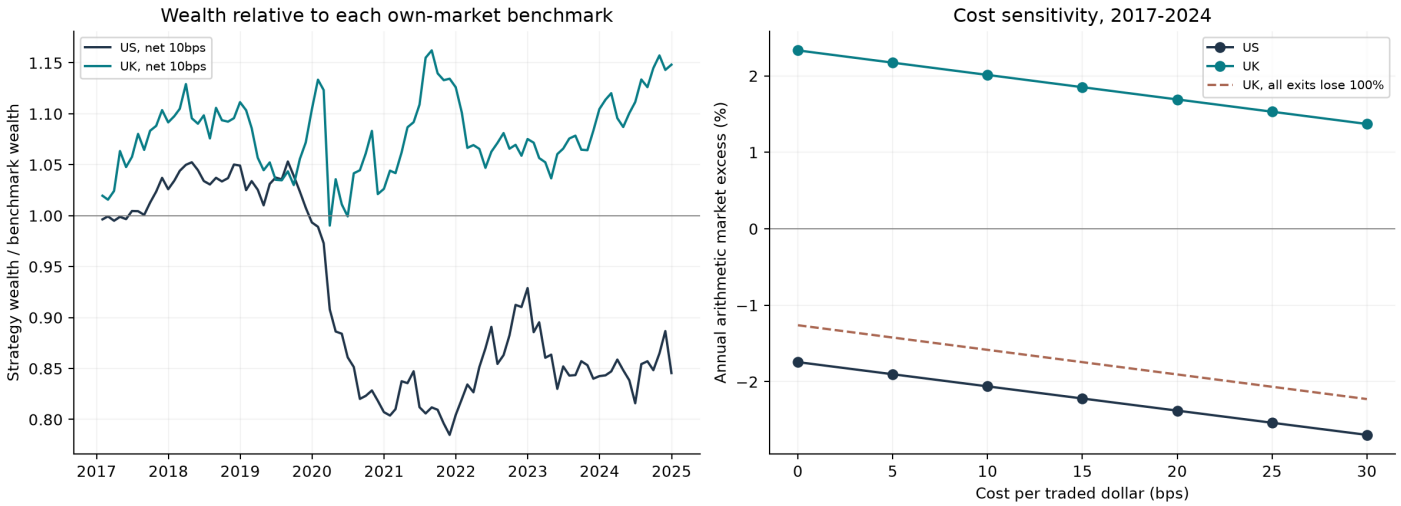
WRDS query 11803424 supplies 7,906,471 daily rows from January 2013 through December 2024 for FIC=GBR. We keep the final available daily observation per issue and calendar month. The analysis keeps EXCHG=194, TPCI=0 and CURCDD=GBP, with the largest eligible issue per company each month. Active and inactive securities remain. The sample contains 178,848 monthly observations across 2,008 issues before history and portfolio screens. No stock-level records appear in the public repository.

Compustat adjusted price is PRCCD / QUNIT / AJEXDI, adjusted shares are CSHOC x AJEXDI, and the total-return index is adjusted price x TRFD. A factor of one replaces TRFD only for issues whose entire observed factor series is missing, following the WRDS no-dividend convention. Month-to-month ratios require consecutive calendar months. Gaps invalidate signal histories. For held positions, missing months carry at zero and remain ineligible for new purchases. Dividend cash uses the same monthly total-return-minus-price-return proxy as the US calculation, multiplied by prior adjusted price. The split-adjusted share-count term remains part of net payout.

First formation is December 2016, followed by 96 investment months. The eligible universe ranges from 961 to 1000 names. We retain the 100-stock baseline, and also report a proportional top-10% holding rule. Quarter-end

signals earn returns only in the next month. The UK comparator uses the same history-qualified largest-1,000 universe, capitalization weights and quarterly timing. This differs from the broad CRSP US index. UK cash earns zero because no UK short-rate series is supplied. No UK Sharpe or factor alpha is claimed.

2017-2024, net 10 bps	Annual return	Own benchmark	Above market / yr
US, USD	10.66%	13.01%	-2.06%
UK base, GBP	7.23%	5.40%	2.01%
UK exit stress, GBP	1.65%	3.55%	-1.58%
UK plus 50bps buy tax	6.34%	5.40%	1.17%



Country performance relative to each own-market comparator. Local currency returns do not represent a common USD investor.

We lack CRSP-equivalent observed UK terminal payments. The base scenario liquidates disappearing issues at the last known value in the next month. A severe stress assigns a 100% loss to every disappearing issue, including those that may have merged for positive consideration. Both the portfolio and its UK comparator use the same scenario. This is a sensitivity range, not an estimate of actual delisting losses.

The generic 0-30 bps scenarios exclude taxes. A separate UK test charges 0.5% on purchase notional, including the initial purchase, in addition to the symmetric trading cost. It assumes no exemptions. HMRC documents the principal electronic share-purchase SDRT rate at 0.5%. The scenario is deliberately explicit rather than treating 5-30 bps as a complete UK implementation cost.

The UK base results are more encouraging in this selected sample. The severe exit stress reverses its market excess, so the apparent advantage depends on terminal-value assumptions. Current country, exchange and issue-type classifications introduce historical-selection uncertainty. UK GBP and US USD returns have different currency exposures. Benchmarks differ, UK delisting payments are incomplete and a UK factor model is absent. This extension follows presentation feedback after the sample was already seen. It does not establish a robust country effect or justify changing the original formula to maximize these eight years of results.

Source methodology: WRDS, Computing Returns. Public metadata in data/uk_source.json and results/uk_comparison/run_meta.json record the export hash, variables and transformations. Licensed source data remain in the private course package.

References

Blitz, D. and van Vliet, P. (2018). The Conservative Formula: Quantitative Investing Made Easy. The Journal of Portfolio Management 44(7). Supplied working paper: SSRN 3145152.

CRSP. Calculations and Index Methodologies. Official methodology guide. WRDS CRSP legacy and Fama-French web exports supply the local inputs; receipts and hashes accompany the private data.

Clarabel. Python Getting Started. Official solver documentation. The assignment is reproduced in Instructions.md. Methods, deviations and independent checks are detailed in the notebook, FINAL_REVIEW.md and results/.

Result provenance: 14 completed studies, 102 files. Data-manifest SHA-256: 4816d582000aecfca4c73ad6... Results last updated UTC: 2026-10-19T01:17:00+00:00.